

J&A AUTOMATION LLC · USA DIVISION

ENTERPRISE PLATFORM ADMINISTRATION & OPERATIONS MANUAL

Complete executive and operational authority for client management, project commercial architecture, master timesheets, field diagnostics, invoice lifecycles, financial intelligence, generated artifacts, and audit compliance.

OWNER & EXECUTIVE ADMIN PRIVILEGE LEVEL · CLIENT ESSENTIAL RELEASE

System Scope & Authority:

J&A Private Operations Monolith
Authority: Client Essential Spec & Checklist (2026-08-24)
Operating Database: Production SQLite with Exact Money

Classification & Governance:

CONFIDENTIAL & PROPRIETARY
Authoritative operations guide for Owners, Directors, and Finance Administrators.
Exhaustive reference of all buttons, inputs, dropdowns, and database mappings.

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Client Essential Production Invariants

Invariant Area	Contractual Mandate (Spec 2026-08-22 / 2026-08-24)	Database & Engine Enforcement
Exact Money Semantics	Floating point math is strictly forbidden. All monetary calculations use exact integer cents.	Stored as integer cents (e.g. subtotal_minor) and computed using deterministic BigInt arithmetic.
Worker Privacy Firewall	Client billing rates (\$70/h, \$55/h) and company margins are strictly hidden from technicians and field staff.	Server-side DTO allowlists; workers only see their own internal agreed wage and settlement statements.
Immutable Invoices	Once an invoice is issued to a client, it becomes an immutable historical snapshot. Editing is blocked.	Cryptographic snapshot stored in invoice.snapshot_json; corrections require formal Credit Notes.
Customer Period Reports	Clients only pay when signed by their representative. Reports must NEVER display monetary figures.	Dedicated zero-money allowlist rendering with formal physical signature block.
Reference Hours vs Actual	Configured reference hours (10h, 12h) and minimum billable rules must NEVER fabricate actual worker hours.	Actual minutes logged on site remain the immutable foundation for compensation and audit.
Safe Entity Deletion	Empty projects or clients created by mistake can be hard-deleted; entities with activity must be archived.	Database guards verify zero time entries, zero expenses, and zero invoices before allowing hard deletion.

1. EXECUTIVE OVERVIEW, ARCHITECTURE & ROLE PRIVILEGE MATRIX

The **J&A Automation Platform** is a modular monolith designed to operate J&A's complete field engineering and industrial controls business without relying on disconnected spreadsheets or manual workarounds.

Complete Navigation & Section Inventory

Section / Hub	URL Route	Worker Access	Owner Access	Operational Scope
Dashboard / Today	/app/today	PERSONAL SHIFT	EXECUTIVE KPIS	Worker logs daily shift; Owner tracks company WIP, overdue invoices, pending approvals.
Clients	/app/clients	BLOCKED	FULL ACCESS	Master client directory, billing addresses, contacts, tax IDs, deletion/archival.
Projects	/app/projects	BLOCKED	FULL ACCESS	Project setup, commercial billing models, decimal shift hours, milestones, team assignments.
Time Tracking	/app/time	OWN TIMESHEET	MASTER OVERSIGHT	Technicians log hours; Owner oversees all site personnel and edits hours with audit logging.
Approvals Queue	/app/approvals	BLOCKED	FULL ACCESS	Batch approval and rejection queue for submitted weekly timesheets and field expenses.
Expenses	/app/expenses	OWN RECEIPTS	COMPANY CONTROL	Worker uploads receipts; Owner reviews, sets reimbursable/markup/all-in status and approves.
Field Reports	/app/reports	DAILY LOGS	CLIENT & INTERNAL	Technicians submit shift logs; Owner generates Customer Period Reports (zero-money sign-off).
Billing & Invoices	/app/billing	BLOCKED	FULL ACCESS	Unbilled candidate streams, draft generation, immutable freeze, PDF/XML issue, voids, credit notes.
Collections Ledger	/app/ledger	BLOCKED	FULL ACCESS	Accounts receivable aging, recording customer bank payments, wire reconciliation.
Finance Overview	/app/finance	BLOCKED	FULL ACCESS	Real-time project gross margins, commercial rate cards, overtime policies, daily minimums.
Monthly Accounting	/app/accounting	BLOCKED	FULL ACCESS	Financial period close locking, multi-tab Excel workbooks, executive PDF packs.
Audit & Compliance	/app/audit	BLOCKED	FULL ACCESS	Append-only forensic audit trails recording actor ID, IP, before/after states, and timestamps.
Profile & Security	/app/profile	SHARED	SHARED	FIDO2 Passkeys, Authenticator TOTP setup, language selection (EN/ES/PT), password resets.

2. SECURE AUTHENTICATION, PASSKEYS & STEP-UP SECURITY (CORE-01)

The platform implements strict server-side Role-Based Access Control (RBAC) with invitation-only user provisioning, session controls, and cryptographic multi-factor authentication (MFA).

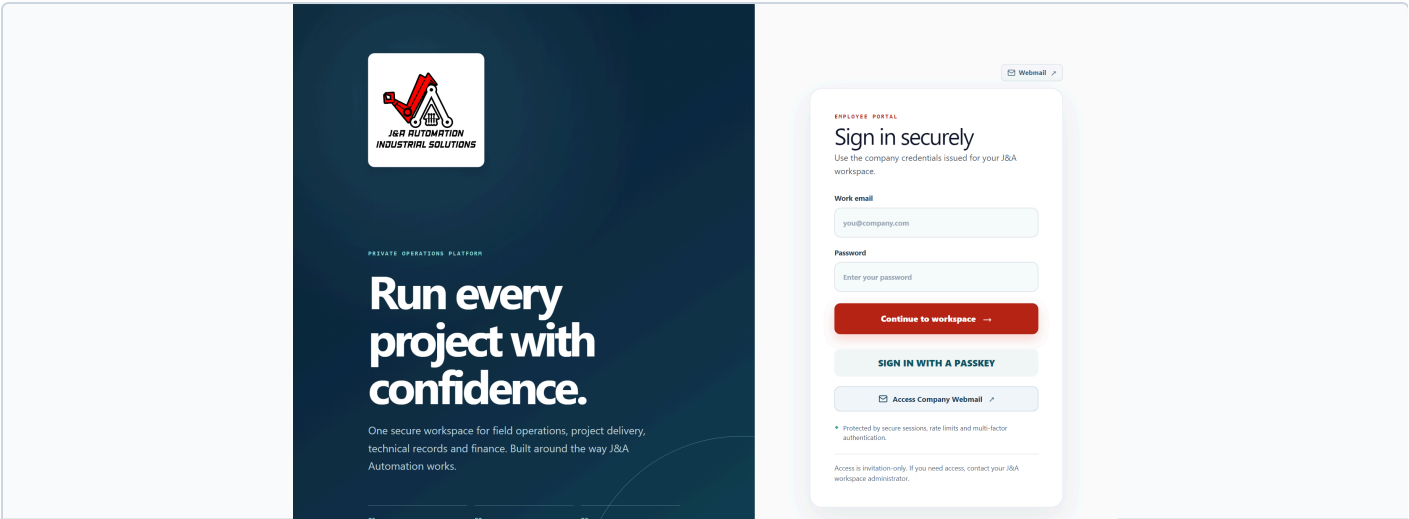


Figure 2.1: Corporate Single Sign-On Portal featuring corporate webmail integration and FIDO2/WebAuthn passkey support.

Authentication Controls, Inputs & Buttons

Interface Element	Type / Tag	Database Target	Function & Security Validation
Email Address	<input type="email">	user.email	Required corporate email address (max 254 chars). Normalized to lowercase.
Password	<input type="password">	user.password_hash	Argon2id cryptographic hash. Minimum 8 characters; never stored in plaintext.
Sign In	Button (POST)	session	Verifies credentials, generates secure session cookie with SameSite=Lax and HttpOnly.
Sign in with Passkey	WebAuthn API	passkey_credential	Hardware-bound biometric login (Windows Hello, Apple TouchID/FaceID, YubiKey). Zero phishing risk.
MFA TOTP Code	<input type="text">	mfa_totp.secret	6-digit RFC 6238 time-based code from Google Authenticator, 1Password, or Microsoft Authenticator.

Step-Up Security Invariant: High-privilege actions (issuing invoices, recording payments, modifying commercial rate cards, or downloading accounting close packs) enforce step-up authentication to protect against session hijacking.

3. CLIENT DIRECTORY: INPUTS, CONTACTS & DELETION LIFECYCLE (CORE-02)

The Client Directory governs corporate accounts, billing addresses, tax numbers, and contact roles for all industrial customers (e.g., IMPC Gmbh, BMW, BBS Automation).

Client Master Data Inputs & Database Columns

Input Box / Dropdown	Field Name	Database Column	Type & Constraints	Operational Definition
Legal Name	legalName	<code>client.legal_name</code>	TEXT (2-200 chars, req)	Full legal company name that appears on formal invoices and tax documents.
Display Name	displayName	<code>client.display_name</code>	TEXT (2-200 chars, req)	Short commercial trading name used in project pickers and daily navigation.
Client Number	clientNumber	<code>client.client_number</code>	TEXT (e.g. CLI-001)	Unique client accounting code used to prefix project numbers (e.g., CLI-001-P-001).
Tax ID / VAT / EIN	taxId	<code>client.tax_id</code>	TEXT (opt)	Corporate tax identification number printed on official billing headers.
Billing Currency	currency	<code>client.currency</code>	SELECT (EUR, USD, etc.)	Default currency for all client projects and invoices. Projects must match client currency.
Headquarters Timezone	timezone	<code>client.timezone</code>	SELECT (IANA Timezone)	Timezone for calendar scheduling and period closes (e.g., Europe/Madrid).
Billing Email	billingEmail	<code>client.billing_email</code>	EMAIL (max 254 chars)	Official Accounts Payable email address where invoices and payment notices are delivered.
Billing Address	billingAddress	<code>client.billing_address</code>	TEXT (max 5000 chars)	Registered corporate address rendered on invoice PDFs under the "Bill To" section.
Payment Terms (Days)	paymentTermsDays	<code>client.payment_terms_days</code>	INTEGER (e.g. 30, 45, 60)	Standard credit terms. Automatically sets invoice due dates upon draft generation.

Interactive Client Buttons & Lifecycle Actions

Button Label	Action Endpoint	Permission	Lifecycle & Database Behavior
+ Create Client	<code>?/createClient</code>	Owner / Finance	Opens modal and inserts client and primary contact in <code>client</code> and <code>client_contact</code> .
Save Client	<code>?/updateClient</code>	Owner / Finance	Updates client master data with optimistic concurrency check (<code>version</code>).
+ Add Contact	<code>?/createContact</code>	Owner / Finance	Adds operational or accounts payable stakeholder with <code>is_billing_contact</code> and <code>is_primary</code> flags.
Archive Client	<code>?/transitionClient</code>	Owner / Finance	Soft-archives client (<code>status = 'archived'</code>); preserves historical invoices and contracts.
Delete Client	<code>?/deleteClient</code>	Owner Only	Hard Deletion: Permanently removes empty clients with 0 projects and 0 invoices. Blocked if activity exists.

4. PROJECTS MASTER & COMMERCIAL BILLING MODELS (CORE-02 & CORE-03)

The Projects Hub serves as the operational center where customer contracts, delivery milestones, budgets, and commercial billing models are configured.

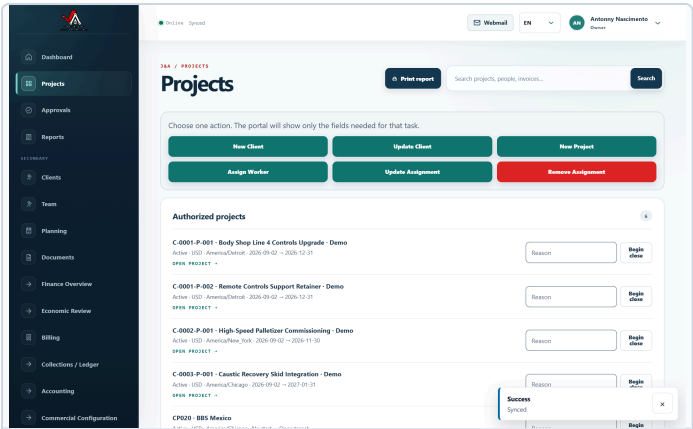


Figure 4.1: Projects Hub displaying active customer contracts, commercial models, and assigned teams.

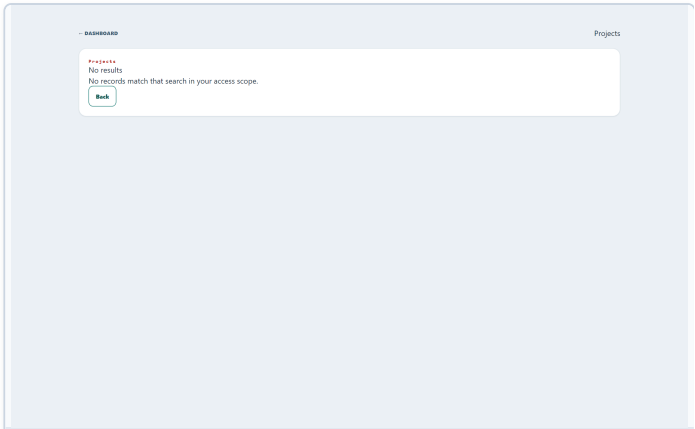


Figure 4.2: Project Detail panel for CP020 BBS Mexico showing PO number, rate cards, and phases.

All 7 Commercial Billing Models Dropdown Explained

Billing Model Option	Database Value	Commercial Definition & Invoicing Logic
Time & Materials (T&M)	tm	Standard hourly billing. Billed as actual hours worked × approved role bill rate (\$70/h, \$55/h).
T&M Daily Minimum	tm_daily_minimum	Hourly billing with guaranteed minimum hours per dispatch day (e.g. minimum 8.0h guaranteed if technician arrives on site).
All-In Package	all_in	Flat daily or weekly package rate covering all technician labor, local travel, and expenses without hourly itemization.
Capped T&M	capped_tm	Hourly billing bounded by a mandatory PO financial ceiling. System halts billing candidates if cap is exceeded.
Fixed Milestone	milestone	Deliverable-based fixed fee (e.g. 50% on design sign-off, 50% on commissioning acceptance). Hours are tracked for internal margin only.
Hybrid	hybrid	Blended contract with fixed milestone deliverables plus separate hourly support rates for on-site commissioning.
Internal / Overhead	internal	Internal engineering, R&D, training, or shop pre-assembly. Non-billable to external clients.

5. DECIMAL HOURS, BUDGETS & PROJECT LIFECYCLE CONTROLS

Project parameters support precise decimal hours, comprehensive budget tracking, and deterministic lifecycle states.

Project Form Inputs, Decimal Hours & Database Columns

Input Box / Dropdown	Field Name	Database Target	Type / Range	Operational Definition
Project Name	name	<code>project.name</code>	TEXT (2-200 chars, req)	Descriptive project title (e.g., <i>BBS Mexico Body Shop Commissioning</i>).
Cost Center Code	costCenterCode	<code>project.cost_center_code</code>	TEXT (1-120 chars, req)	Required accounting identifier (e.g., CP020) linking all project costs to general ledger.
PO / Reference	poNumber	<code>project.po_number</code>	TEXT (max 200 chars)	Customer purchase order reference. Automatically printed on invoices and period reports.
Expected Hours / Day	expectedHoursPerDay	<code>project.expected_minutes_per_day</code>	NUMBER (step 0.25, 0-24, default 10.0)	Reference Shift Hours: Standard working day duration (e.g. 10.0 or 8.5). Converted to exact integer minutes in SQLite (hours × 60). Never fabricates unworked time.
Client Minimum Hours	clientDailyMinimumHours	<code>project.client_daily_minimum_minutes</code>	NUMBER (step 0.25, 0-24, opt)	Minimum Billable Hours: Minimum hours charged to client per dispatch day (e.g. 8.0). Stored as exact integer minutes.
Budget Type	budgetType	<code>project.budget_type</code>	SELECT (6 options)	Options: none, revenue, purchase_order, labor, travel, combined. Controls project budget threshold warnings.
Revenue Budget	revenueBudgetMinor	<code>project.revenue_budget_minor</code>	MONEY (cents, BigInt)	Maximum projected revenue. Stored as integer cents (e.g., \$50,000.00 = 5000000).
Weekly Close Enabled	weeklyCloseEnabled	<code>project.weekly_close_enabled</code>	CHECKBOX (0 or 1)	When checked, locks timesheets every Sunday midnight to prevent retroactive technician edits.
Daily Report Required	dailyReportRequired	<code>project.daily_report_required</code>	CHECKBOX (0 or 1)	Enforces mandatory submission of daily field reports before weekly timesheets can be approved.

Project Buttons & Lifecycle State Transitions

Button Label	Action Endpoint	Allowed Initial State	Target State & Database Behavior
+ Create Project	?/createProject	New Modal	Inserts project row, creates base schedule (Mon-Sat expected minutes, Sun 0), logs audit.
Save Project	?/updateProject	Any active state	Updates project fields with optimistic concurrency check; validates decimal hours between 0 and 24.
Begin Close	?/transitionProject	active, paused	Moves status to closing. Blocks new worker assignments while allowing final timesheets to settle.
Close Project	?/transitionProject	closing	Moves status to closed. Records official <code>actual_end_date</code> in database.
Archive Project	?/transitionProject	closed	Soft-archives project (<code>status = 'archived'</code>). Hides from operational pickers but retains full financial history.
Restore Project	?/transitionProject	archived	Restores project back to active operational status.
Delete Project	?/deleteProject	active (draft)	Hard Deletion: Permanently removes empty projects with 0 hours, 0 expenses, and 0 invoices.

6. RATE CARDS, OVERTIME POLICIES & WORKFORCE ASSIGNMENTS (CORE-03 & CORE-05)

The platform supports J&A's specialized commercial rules that dictate how actual floor hours translate into client invoices and internal worker compensation.

1. Standard Corporate Rate Card Matrix

Engineering Specialization	Client Billing Rate	Overtime Threshold	Overtime Treatment
Lead PLC Engineer (Gabriel Santos)	\$70.00 / hour	After 10h/day or 60h/week	Proportional multiplier (1.5x) when contract is Time & Materials.
Automation Specialist (Maico Silva)	\$55.00 / hour	After 10h/day or 60h/week	Proportional multiplier (1.5x) or agreed contract rate.
Robotics Programmer (Victor Lima)	\$55.00 / hour	After 10h/day or 60h/week	Separated on billing candidate streams.
Commissioning Tech (Andrew Miller)	\$55.00 / hour	After 10h/day or 60h/week	Separated on billing candidate streams.

2. Workforce Assignment Inputs & Controls

Assignment Field	Database Target	Type / Format	Operational Definition
Technician	project_member.user_id	SELECT (Active Workers)	Selects engineer to assign to the project roster.
Role on Project	project_member.role_on_project	SELECT (Roles)	Lead PLC, Robotics Programmer, Automation Specialist, Commissioning Tech.
Starts On / Ends On	project_member.starts_on	DATE (ISO format)	Assignment validity timeframe. Workers can only log hours within their active assignment window.
Client Bill Rate Override	assignment_rate_override.client_bill_rate	MONEY (cents, opt)	Optional project-specific client billing rate overriding default rate card.

3. Assignment Buttons

+ Assign Worker : Enrolls engineer in project team with date bounds. **End Assignment** : Gracefully terminates assignment, preventing future time logs while preserving historical logs.

7. MASTER TIME TRACKING & OPERATIONAL OVERSIGHT (CORE-04)

The Master Time Tracking interface enables executive oversight across all active personnel on site, comparing planned shifts against actual logged hours.

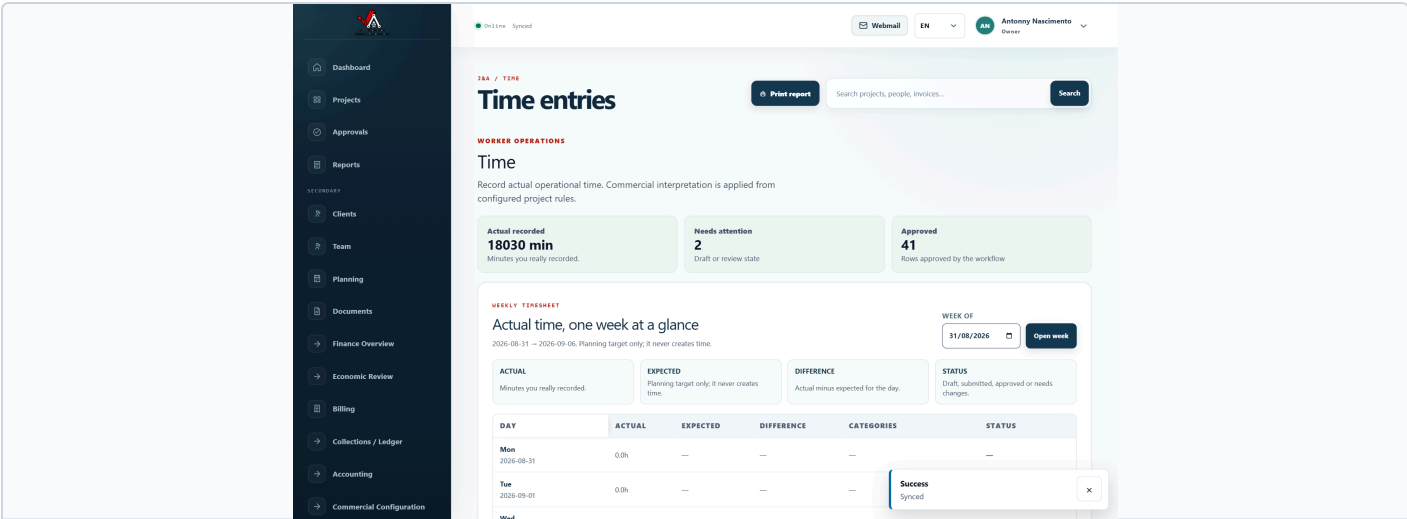


Figure 7.1: Master Time Tracking calendar showing weekly logs (CW31/CW32) for all field workers across all projects.

Master Time Tracking Controls & Input Fields

Input / Selector	Database Target	Options / Format	Operational Meaning
Work Date	time_entry.work_date	DATE (YYYY-MM-DD)	Day shift was performed on the factory floor.
Project Picker	time_entry.project_id	SELECT (Active Projects)	Project against which labor is recorded. Owners can log/edit for any active project.
Actual Hours	time_entry.duration_minutes	NUMBER (step 0.25, decimal)	Exact hours worked on site. Stored in SQLite as exact integer minutes (hours × 60).
Work Category	time_entry.category	SELECT (4 categories)	Options: regular (standard work), overtime (excess hours), travel (transit), standby (plant delay).
Task Description	time_entry.description	TEXT (max 5000 chars)	Detailed engineering summary (e.g. "Line 1 Safety circuit integration and interlock testing").

Time Tracking Action Buttons

Save Entry : Saves time entry as draft. **Submit Week** : Submits weekly sheet for approval. **Copy Prior Week** : Clones prior week project rows with 0.0h. **Delete Entry** : Removes draft entry.

8. APPROVALS QUEUE: TIMESHEET & EXPENSE WORKFLOWS (CORE-08)

The Approvals Hub functions as the quality control firewall between technician logs and client-facing billing candidate pools.

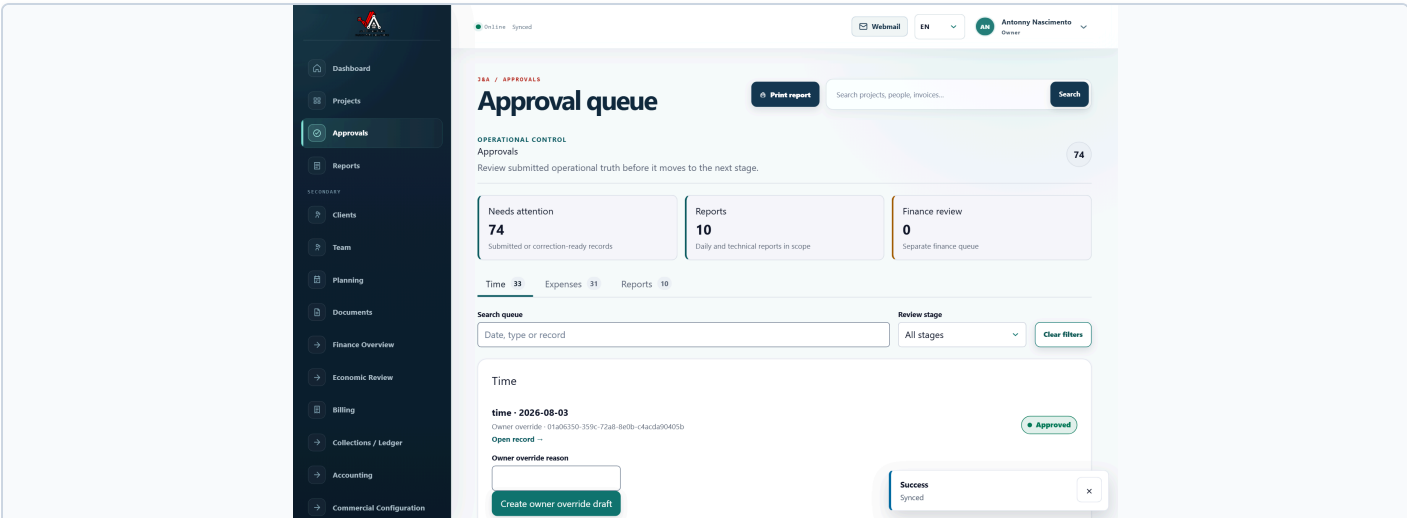


Figure 8.1: Approvals Queue allowing single-click and batch verification of hours, daily reports, and expense receipts.

Timesheet & Expense Lifecycle States

State Badge	Database Status	Worker Permitted Actions	Owner / PM Actions
Draft	draft	Edit hours, change notes, delete entry.	View only.
Submitted	submitted	Locked. Cannot edit or delete.	Approve entry or Reject with required feedback reason.
Approved	approved	Locked. Displayed on compensation statement.	Queued into unbilled candidate streams for invoice generation.
Rejected	rejected	Unlocked with rejection notes; worker can revise.	Awaiting revised re-submission.

Approvals Buttons & Action Endpoints

Button Label	Action Endpoint	Required Inputs	Database Effect
Approve Week	*/approveTimeWeek	projectId, userId, weekEnding	Transitions all entries from submitted to approved. Unlocks billing candidate streams.
Reject Week	*/rejectTimeWeek	reason (TEXT, mandatory)	Sets status to rejected, logs rejection reason in time_entry.rejection_reason, notifies worker.
Approve Expense	*/approveExpense	expenseId, billingTreatment	Clears expense for settlement payout and adds to billable expense stream if reimbursable.
Reject Expense	*/rejectExpense	reason (TEXT, mandatory)	Rejects expense item, requiring worker to re-upload clear receipt or fix amount.

9. FIELD OPERATIONS & GENERATED PLC TECHNICAL REPORTS (CORE-07)

J&A Automation field services combine daily operational shift records with deep technical PLC program change logs.

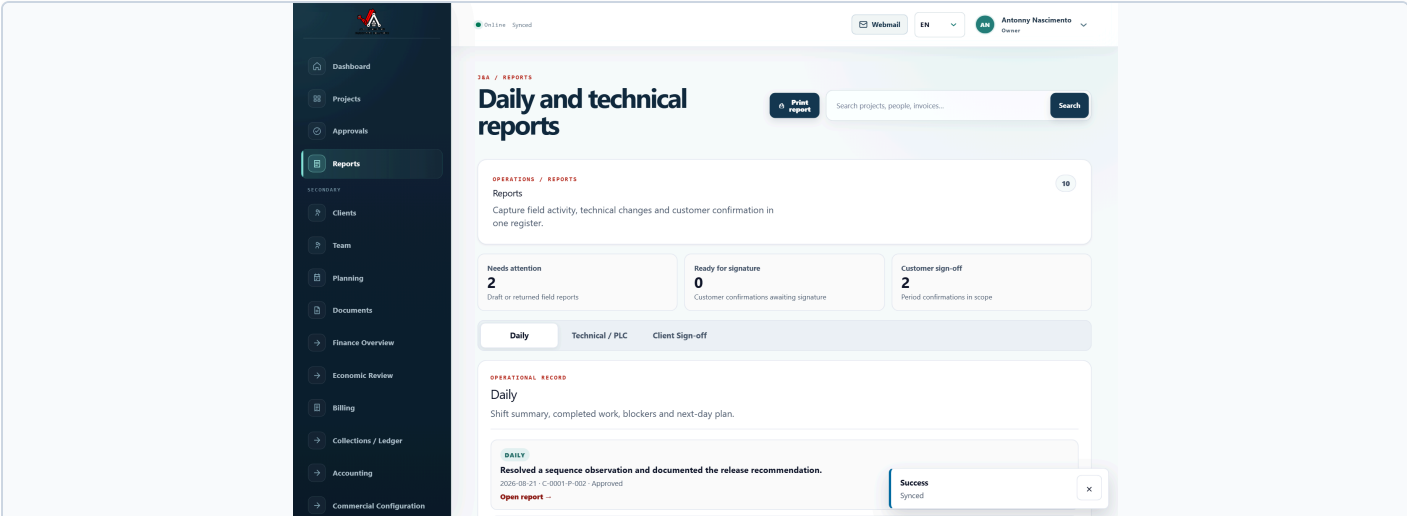


Figure 9.1: Daily Field Reports Hub showing shifts, machines worked, safety checks, and customer sign-off readiness.

Generated Technical PLC Diagnostics PDF

When engineers execute programming or safety modifications on customer equipment (e.g. Siemens S7-1500 or Rockwell GuardLogix), the platform compiles an official diagnostic artifact:



Figure 9.2: Official Generated Technical PLC Report (PDF) detailing system ID, Siemens CPU 1517F-3, root cause diagnosis, safety SIL3 test results, and backup hash.

Technical Report Form Inputs & Database Mappings

Report Input	Database Target	Format	Operational Scope
Equipment / Station	<code>technical_report.equipment_name</code>	TEXT	Machine name or production line station (e.g., <i>Station 040 Welder</i>).
Controller / Hardware	<code>technical_report.controller_model</code>	TEXT	PLC CPU model (e.g., <i>Siemens CPU 1517F-3 PN/DP, AB ControlLogix 5580</i>).
Root Cause Diagnosis	<code>technical_report.root_cause</code>	TEXT	Engineering explanation of the defect, timing fault, or sensor malfunction.
Safety SIL3 Check	<code>technical_report.safety_check</code>	BOOLEAN	Confirms that safety interlocks, light curtains, and E-stops were tested and validated.
Program Backup File	<code>technical_report.backup_hash</code>	FILE UPLOAD	Stores PLC project archive (.zap18, .acd) with SHA-256 integrity checksum.

10. CUSTOMER SIGN-OFF GATE & ZERO-MONEY PERIOD REPORTS (CORE-07 & CORE-13)

A central contractual requirement of J&A Automation operations is the **Customer Sign-off Gate**: *"solo pagan si tiene la firma, no puede tener valores de dinero ni del contratado ni del trabajador, eso hay que generar para la firma del cliente"*.*.

Customer Sign-off Report vs. Internal Financial Report

The platform generates two distinct period reports from the exact same source hours:



Customer vs. Internal Report Comparison Matrix

Report Feature	Customer Period Report (Client Sign-Off)	Internal Financial Period Report (Owner Admin)
Monetary Figures (\$ / €)	STRICTLY STRIPPED (ZERO MONEY)	FULL DISCLOSURE (RATES, COSTS, MARGINS)
Worker Internal Pay	Hidden (Privacy firewall enforced)	Visible (\$40/h rate, total worker compensation)
Client Billing Rates	Hidden (Zero commercial rates)	Visible (\$70/h billing rate, total billable amount)
Signature Block	Formal physical sign-off box for Plant Representative	Internal sign-off / accounting audit approval
Pre-Billing Invoicing Gate	Mandatory prerequisite for final invoice issuance	Used for internal profit reconciliation

11. EXPENSES HUB & COMMERCIAL BILLING TREATMENTS (CORE-06)

The Expenses Hub manages all site travel, flights, lodging, rental cars, and hardware purchases incurred during commissioning.

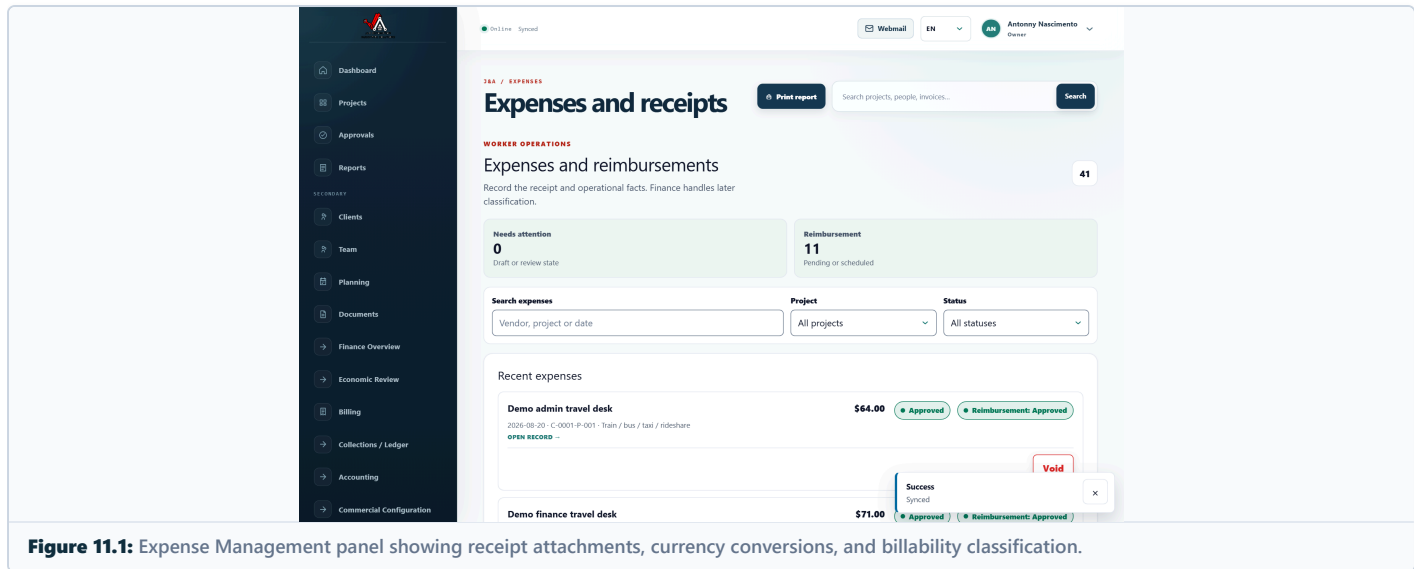


Figure 11.1: Expense Management panel showing receipt attachments, currency conversions, and billability classification.

Expense Input Fields, Categories & Database Columns

Input Box / Dropdown	Field Name	Database Target	Available Options / Rules
Incurred On	incurredOn	expense.incurred_on	DATE (YYYY-MM-DD, req) - Transaction date matching the physical receipt.
Project	projectId	expense.project_id	SELECT (Active Projects) - Assigns cost to project cost center.
Expense Category	category	expense.category	SELECT: lodging (hotel), meals (food/per diem), travel_transit (flights, trains, rental cars), mileage (vehicle usage), materials_tools (site parts), other.
Amount	amount	expense.amount_minor	NUMBER (decimal, 2 places). Converted and stored as exact integer cents.
Currency	currency	expense.currency	SELECT: USD, EUR, MXN, CAD. Converted to project currency using verified exchange rates.
Merchant / Vendor	merchant	expense.merchant	TEXT (e.g., <i>Marriott Spartanburg, Hertz DFW, Delta Air Lines</i>).
Receipt File	receiptFile	expense.receipt_storage_key	FILE UPLOAD (PDF, PNG, JPEG, WebP, max 50 MB). Encrypted in private artifact storage.
Billing Treatment	billingTreatment	expense.billing_treatment	SELECT (Owner Only): reimbursable_at_cost (1:1 client billing), reimbursable_with_markup, all_in_included (absorbed in project fee), non_billable.

12. BILLING HUB, CANDIDATE STREAMS & GENERATED INVOICE PDFS (CORE-10 & CORE-11)

The Billing Hub orchestrates invoice generation from approved hours and expenses across all active project streams.

Figure 12.1: Billing Hub showing invoice statuses (Draft, Issued, Paid), service periods, and generation actions.

Generated Labor vs. Expenses Invoice Artifacts

Labor and Expenses generate distinct official PDF documents with independent tax profiles and accounting numbers:

Figure 12.2: Generated Labor Invoice (CP020-013) with 4-column layout, boxed totals (\$57,437.25), and wire terms.

Figure 12.3: Generated Expenses Invoice (CP020-014) itemizing Marriott, Hertz, American Airlines, and fuel (\$2,197.00).

Three Independent Candidate Streams (Duplicate Billing Prevention)

- **Labor Candidate Stream:** Approved timesheet hours grouped by technician and role rate card.
- **Expenses Candidate Stream:** Approved reimbursable expenses with receipt attachments.
- **Milestones Candidate Stream:** Completed project milestones ready for commercial billing.

13. INVOICE PREVIEW, BOXED TOTALS & WELLS FARGO REMITTANCE

Every invoice generated by the system presents a standardized, publication-grade corporate layout with draft customizations:

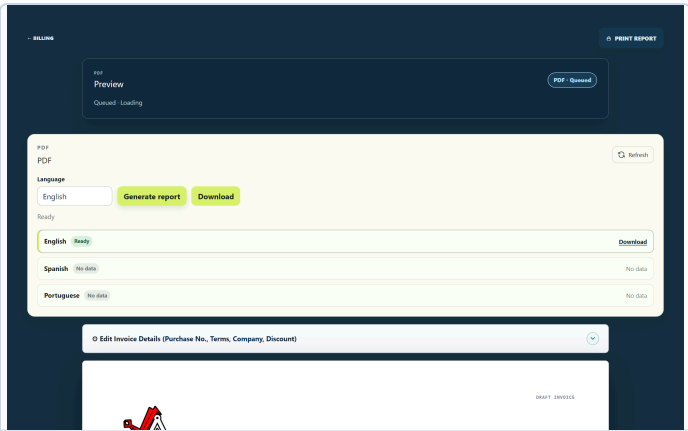


Figure 13.1: Web Invoice Preview showing 4 columns, boxed totals, and Wells Fargo wire instructions.

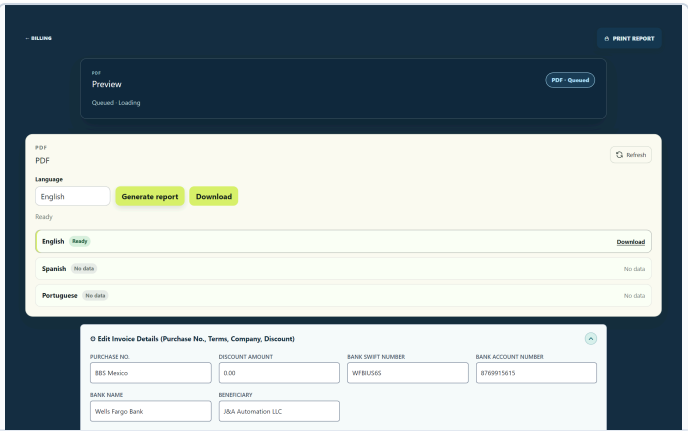


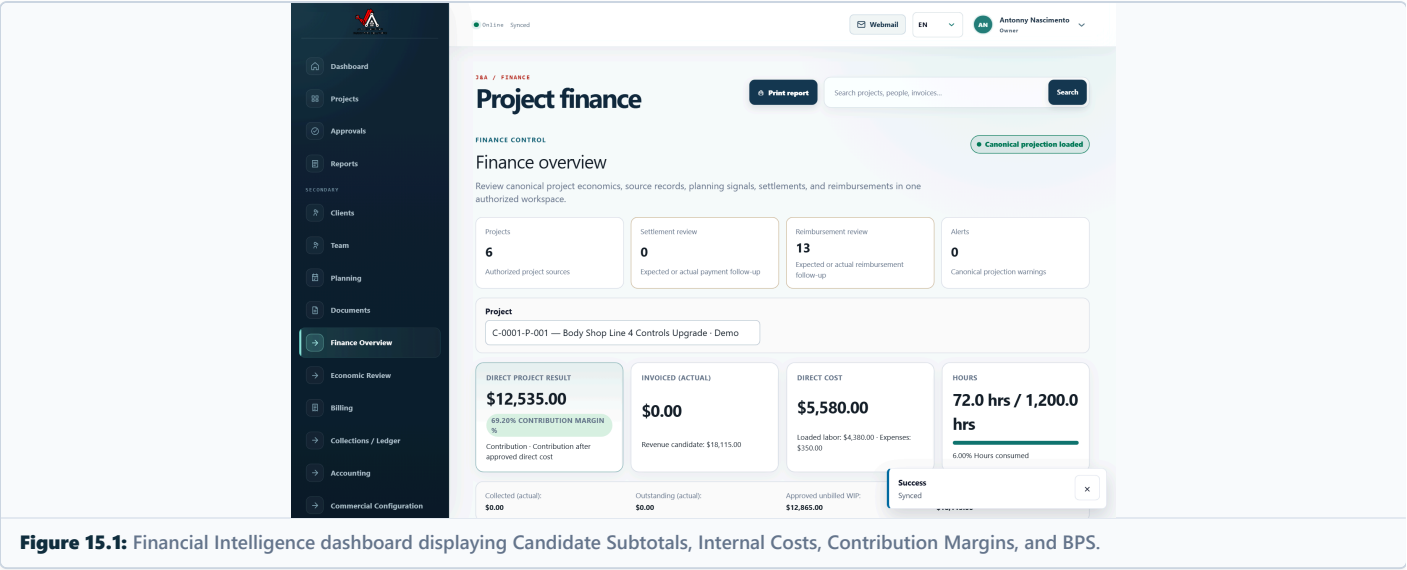
Figure 13.2: Draft Customization Accordion allowing direct editing of Purchase No., discounts, and banking details.

Invoice Draft Inputs & Actions

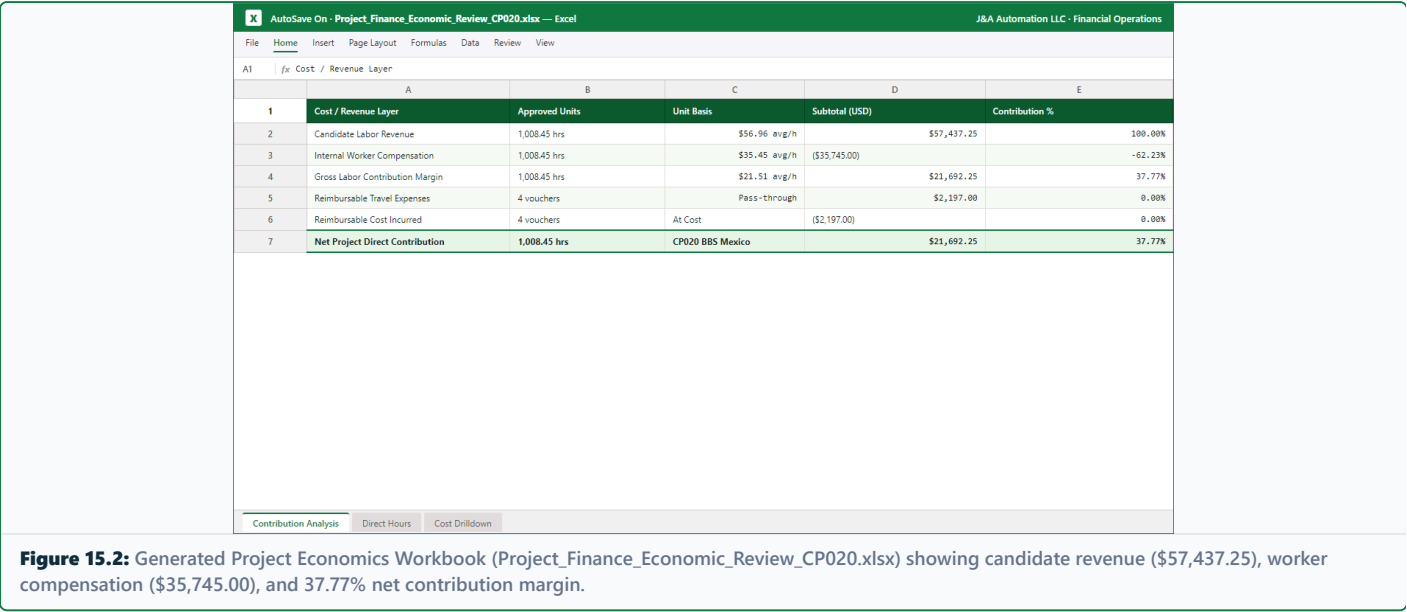
Button / Field	Target	Behavior & Purpose
Generate Draft Invoice	?/generateInvoiceDraft	Collects unbilled candidate stream rows into an editable draft invoice.
Purchase No. (PO Number)	invoice.po_number	Editable customer PO reference displayed prominently on the invoice header.
Banking Details	invoice.banking_details	Remittance info: Wells Fargo Bank, Swift: WFBUIUS6S, Account: 8769915615.
Finalize & Issue	?/issueInvoice	Cryptographic Freeze: Permanently locks source timesheets/expenses, stores immutable snapshot in invoice.snapshot_json, and generates official PDF/XML.
Void Invoice	?/voidInvoice	Cancels draft or unpaid invoice with mandatory audit justification reason.

15. FINANCIAL INTELLIGENCE & GENERATED ECONOMICS EXCEL (CORE-09)

The Finance Hub gives executive leadership real-time visibility into project economic viability, gross margins, and labor realization rates.



Generated Project Economics Excel Workbook (.xlsx)



Key Profitability Metrics Explained:

- **Direct Contribution Margin:** Revenue – Direct Labor Costs – Direct Travel Expenses.
- **Basis Points (BPS):** Margin percentage expressed in financial basis points (e.g., 37.77% = 3,777 BPS).
- **Realization Rate:** Effective net revenue earned per technician hour on site.

16. MONTHLY ACCOUNTING CLOSE & MULTI-TAB EXCEL WORKBOOK (CORE-13)

At the close of each calendar month, administrators generate the official **Accounting Pack** to lock the financial period and reconcile revenue, labor costs, and vendor disbursements.

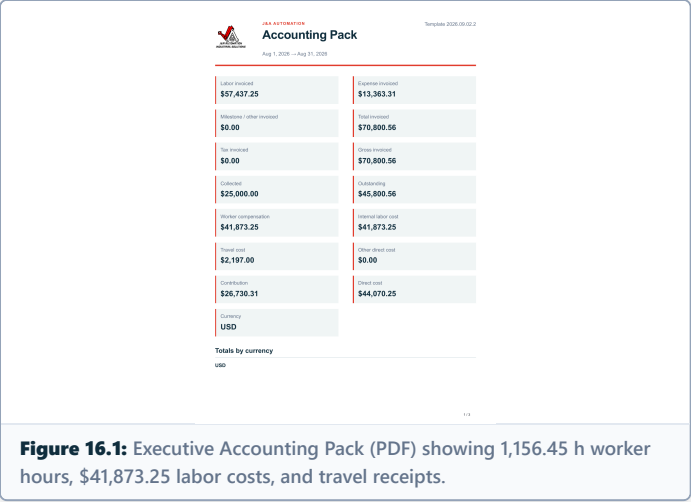


Figure 16.1: Executive Accounting Pack (PDF) showing 1,156.45 h worker hours, \$41,873.25 labor costs, and travel receipts.

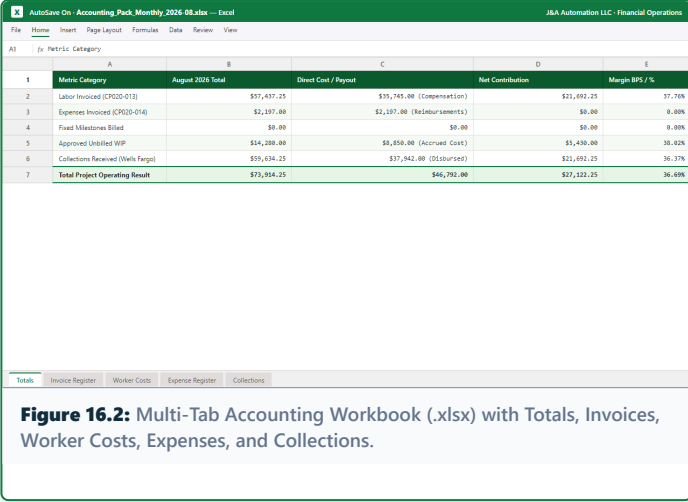


Figure 16.2: Multi-Tab Accounting Workbook (.xlsx) with Totals, Invoices, Worker Costs, Expenses, and Collections.

Accounting Close Buttons & Period Locking

Button Label	Action Endpoint	Operational Effect
Close Accounting Period	?/closeAccountingPeriod	Hard Freeze: Locks month; prevents creation or modification of timesheets, expenses, or invoices in the period.
Generate Accounting Pack	?/generateAccountingPack	Compiles authoritative multi-sheet Excel workbook and executive PDF summary pack.

17. IMMUTABLE SYSTEM AUDIT LOGS & FORENSICS (CORE-15)

The platform maintains an append-only audit trail recording every state change, user authentication, invoice generation, payment recording, and entity deletion.

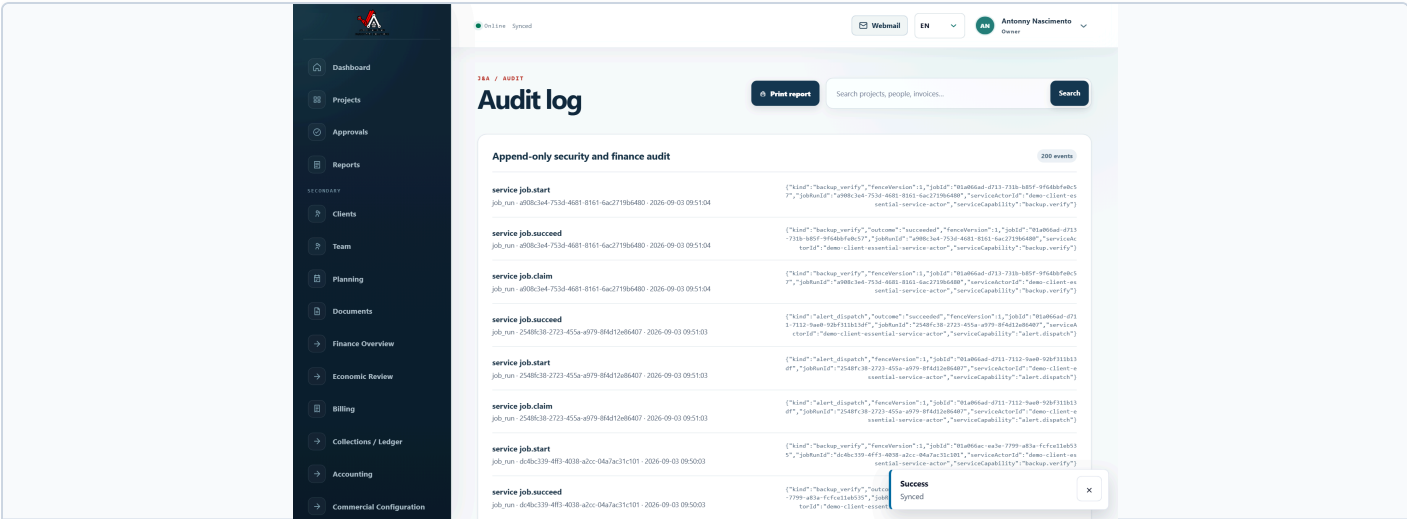


Figure 17.1: System Audit Log recording immutable actor IDs, timestamps, IP addresses, and state changes.

Audit Trail Table Schema & Fields

Audit Field	Database Column	Audit Verification Function
Timestamp	audit_event.created_at	ISO-8601 UTC timestamp recorded at transaction commit time.
Actor ID	audit_event.user_id	User ID and corporate email of the principal who executed the action.
Action Type	audit_event.action	Strictly validated against audit_action_registry (e.g., lifecycle.transition, invoice.issued).
Entity Target	audit_event.entity_type / entity_id	Target table and row identifier (e.g. project, invoice, client).
State Transition	audit_event.payload	JSON payload capturing fromState, toState, modified fields, and mandatory user justification notes.